Estimation of seasonal variations requires that the time series data are recorded at even intervals such as quarterly, monthly, weekly or daily, depending on the nature of the time series. Changes due to seasons, weather conditions and social customs are the primary causes of seasonal variations. The main objective of the measurement of seasonal variation is to study their effect and isolate them from the trend.

Methods of constructing seasonal indices

There are four methods of constructing seasonal indices.

1. Simple averages method
2. Ratio to trend method
3. Percentage moving average method
4. Link relatives method

Among these, we shall discuss the construction of seasonal index by the first method only.

7.3.5 Simple Averages Method

Under this method, the time series data for each of the 4 seasons (for quarterly data) of a particular year are expressed as percentages to the seasonal average for that year.

The percentages for different seasons are averaged over the years by using simple average. The resulting percentages for each of the 4 seasons then constitute the required seasonal indices.

Method of calculating seasonal indices

i) The data is arranged season-wise

ii) The data for all the 4 seasons are added first for all the years and the seasonal averages for each year is computed.

iii) The average of seasonal averages is calculated
(*i.e.*, Grand average = Total of seasonal averages /number of years).

iv) The seasonal average for each year is divided by the corresponding grand average and the results are expressed in percentages and these are called seasonal indices.

Example 7.9

Calculate the seasonal indices for the rain fall (in mm) data in Tamil Nadu given below by simple average method

Year	Season			
	I	II	III	IV
2001	118.4	260.0	379.4	70
2002	85.8	185.4	407.1	8.7
2003	129.8	336.5	403.1	12.0
2004	283.4	360.7	472.1	14.3
2005	231.7	308.5	828.8	15.9

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